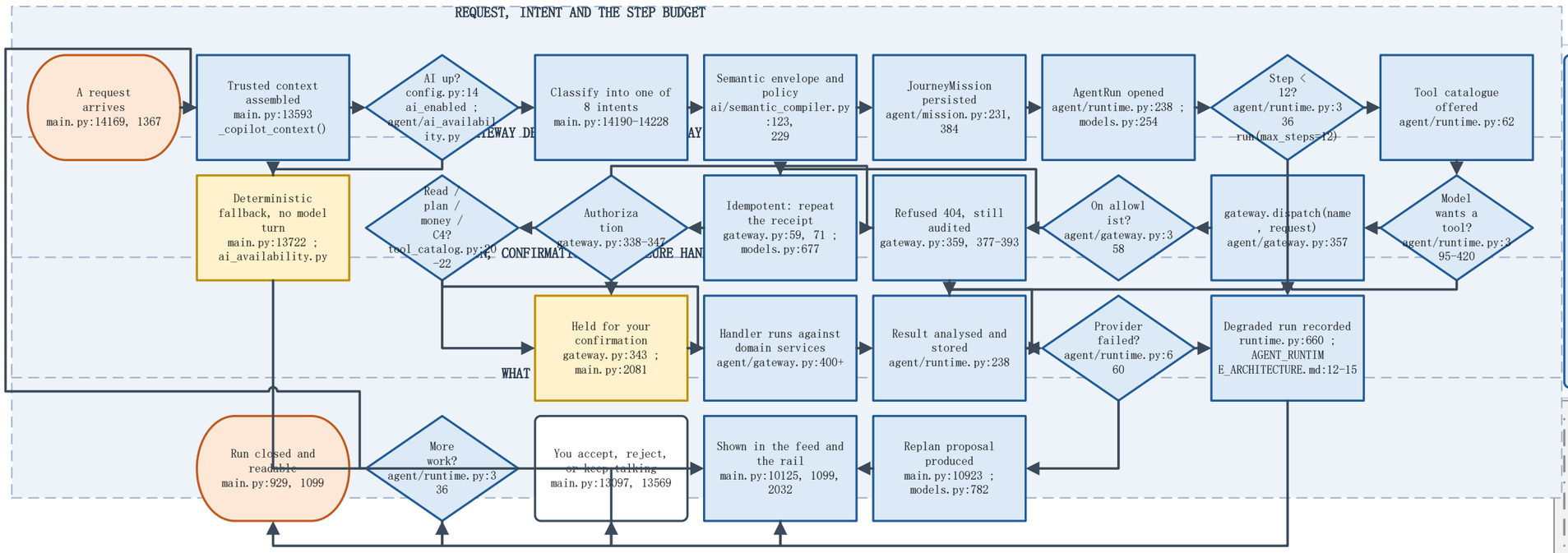
CityNexus | AI Agent Autonomy and Tool-Calling Workflow

CNX-SE-VIS-03 Agent workflow (tool loop and authorization gates)
Rev A • generated 2026-10-22 • source tree D:\moodle\CityNexus @ review/boli-party-occupancy-scope-r1

The path from a sentence to an action, and the three places where the system is allowed to say no. Reads are free; plan changes become proposals; anything touching money or a protected commitment waits for the traveler. A failed provider is recorded as a degraded run instead of quietly rewriting the journey.



WHY THIS IS THE DIAGRAM TO PRESENT: an information page answers a question; this loop is what makes CityNexus an agent application. The model never writes journey state itself - agent/runtime.py:336 asks, agent/gateway.py:357 decides, and a domain service performs.

THREE DOORS, NOT ONE. allowlist (gateway.py:358) then idempotency (gateway.py:59, 71) then authorization (gateway.py:338-347). The PLAN_ONLY boundary is enforced server-side, so a request body claiming to be Hermes cannot widen it. max_steps=12 is a real bound in runtime.py:336; a run that spends it is reported as degraded, not retried invisibly.

The two panels on the right are part of the model, not decoration: the coursework asks to separate what the code does today from what is planned, and claiming a closed automation loop here would be false.

Legend

- Implemented + evidence
- Implemented, not proven on the line
- Missing or future work
- Agent-owned step
- Human in the loop